DIF 1208

El método de sustitución simple. "Opuesto de la regla de la cadena".

Motivación. $f(x) = (x^3+1)^2$

$f'(x) = 2(x^3+1)^2(3x^2)$

$\therefore \int 2(x^3+1)^2 \cdot (3x^2)\,dx = (x^3+1)^2 + C_1$

Ejemplo 1:

$$\int 2x\sqrt{x^2+5}\,dx = \int 2x(x^2+5)^{\frac{1}{2}}\,dx \qquad u = x^2+5 \quad du = 2x\,dx$$

$$= \int \sqrt{u}\,du = \int (u)^{\frac{1}{2}}\,du = \frac{2}{3}u^{\frac{3}{2}} + C_1$$

$$= \frac{2}{3}(x^2+5)^{\frac{3}{2}} + C_1$$

Ejemplo 2:

$u = \ln(x)$

$du = \frac{1}{x}dx$

$$\int \frac{1}{x\ln(x)}\,dx = \int \frac{1}{u}\,du = \ln(|u|) + C$$

$$= \ln(\ln(x)) + C$$

Ejemplo 3:

$$\int x^2\sqrt{x^3-7}\,dx = \int x^2(x^3-7)^{\frac{1}{4}}$$

$u = x^3 \cancel{-7}$

$du = 3x^2\,dx$

$\frac{du}{3} = x^2\,dx$

$$= \int \sqrt[4]{u}\,\frac{du}{3} = \frac{1}{3}\int u^{\frac{1}{4}}\,du = \frac{1}{3}\cdot\frac{4}{5}(u)^{\frac{5}{4}} = \frac{4}{15}\cdot u^{\frac{5}{4}}$$

Ejemplo 4:

$$\int \frac{1}{x(\ln(x))^2} = \int \frac{du}{u^2} = \int u^{-2}\,du = -u^{-1}$$

$$= -(\ln(x))^{-1} + C_1$$

Ejemplo 5:

$$\int \frac{x^5}{x^6+1}\,dx$$

$u = x^6+1$

$du = 6x^5\,dx$

$\frac{du}{6} = x^5\,dx$

$$= \int \frac{\frac{du}{6}}{u} = \frac{1}{6}\int \frac{du}{u} = \frac{1}{6}\ln|u| + C_1 = \frac{1}{6}\ln(x^6+1) + C$$

m) $\int \frac{x}{\sqrt{1+x^2}} dx = \int \frac{du/2}{(u)^{1/2}} = \cancel{\int (u)^{-1/2} du = 2u^{1/2} = 2(x^2+1)^{1/2}}$

$\cancel{2 \cdot 2x(x^2+1)^{\frac{1}{2}}}$

$u = x^2 + 1$

$du = 2x\,dx$

$\frac{du}{2} = x\,dx$

$= \int (u)^{-1/2} \frac{du}{2}$

$= \frac{1}{2} \int (u)^{-1/2} du = \frac{1}{2} 2 (u)^{1/2} = \sqrt{u} + C = \sqrt{x^2+1} + C$

k) $\int (2x-1) e^{x^2-x} = \int e^u du = e^u + C = e^{x^2-x} + C$

$\sqrt{x}' = \frac{1}{2\sqrt{x}}$

$u = x^2 - x$

$du = 2x - 1\,dx$

q) $\int \frac{e^x}{e^x+1} = \int \frac{du}{u} = \int u^{-1} du = \int \frac{1}{u} du = \ln|u| + C$

$= \ln|e^x+1| + C$

$= \ln(e^x+1) + C$

$u = e^x + 1$

$du = e^x dx$

Simplificar as integrais

exemplo:

$u = 4x+5$ $\quad \frac{du}{4} = dx$

$du = 4dx$

$\int \cos(4x+5) dx = \int \cos(u) \frac{du}{4} = \frac{1}{4} \int \cos(u) du$

$= \frac{1}{4} \operatorname{sen}(u) + C = \frac{1}{4} \operatorname{sen}(4x+5) + C$

$\int e^{5x-1} dx = \int e^u \frac{du}{5} = \frac{1}{5} \int e^u du$

$= \frac{1}{5} e^u$

$= \frac{1}{5} e^{5x-1}$

1) $\int \frac{1}{\sqrt{2x+3}}dx = \int \frac{1}{\sqrt{u}} \frac{du}{2} = \frac{1}{2}\int \frac{1}{\sqrt{u}} du = \frac{1}{2}\int u^{-1/2} du$

$u = 2x+3$

$du = 2\,dx$

$\frac{du}{2} = dx$

$= \frac{1}{2} 2 u^{1/2} + C$

$= u^{1/2} + C = \sqrt{2x+3} + C$

$\int 8x^2 \tan(4x^3)dx = 2\int 4x^2 \tan(4x^3)\,dx = 2\int \tan(u)\frac{du}{3}$

$u = 4x^3$

$du = 4\cdot 3x^2 dx$

$\frac{du}{3} = 4x^2 dx$

$= \frac{2}{3}\int \tan(u)\,du$

$= \frac{2}{3}\ln|\sec(u)| + C$

$= \frac{2}{3}\ln|\sec(4x^3)| + C$

$= -\frac{2}{3}\ln|\cos(4x^3)| + C$

Additional:

$\int \cancel{x}(x+7)^{10}dx = \int (u)^{10}du(u-7)$

$u = x+7$

$du = dx$

$= \int (u^{11} - 7u^{10})\,du$

$= \frac{u^{12}}{12} - 7\frac{u^{11}}{11} + C$

$= \frac{(x+7)^{12}}{12} - 7\frac{(x+7)^{11}}{11} + C$

$\int$

г) $\int x^5 \cdot \sqrt{2 + x^3}\, dx$

$u = x^3 + 2 \;|\; -2$

$u - 2 = x^3$

$du = 3x^2\, dx$

$\frac{du}{3} = x^2\, dx \;|\; \cdot x^3$

$\frac{du\, x^3}{3} = x^5\, dx$

$u\sqrt{}$

$u \cdot u^{1/2}$

$$= \int \frac{x^3}{3} \cdot \sqrt{u}\, du$$

$$= \frac{1}{3} \int u - 2 \sqrt{u}\, du$$

$$= \frac{1}{3} \int u\sqrt{u} - 2\sqrt{u}\, du$$

$$= \frac{1}{3} \int u^{3/2} - 2u^{1/2}$$

$$= \frac{1}{3} \left(\frac{u^{5/2}}{5/2} - 2\frac{u^{3/2}}{3/2} \right) =$$

$$\cancel{= \frac{2}{15} u^{5/2} - \frac{4}{9} u^{3/2}}$$

$$= \frac{2}{15} (x^3+2)^{5/2} - \frac{4}{9} (x^3+2)^{3/2}$$